

Ahmed Ikram | ML & AI Engineering

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PROFILE

3rd-year BSc Computer Science (Data Science & AI) student at the University of Dundee, on track for a First-Class degree. Specialises in end-to-end ML pipeline engineering, MLOps infrastructure, and LLM-powered application development - having independently built an anomaly detection engine achieving $99.8\% \pm 0.1\%$ StratifiedKFold CV accuracy and an Agentic RAG assistant cutting inference latency from 25-50s to 10-20s, with all training runs versioned in MLflow on AWS. Applies ML rigorously: leakage-safe feature engineering, cross-validated benchmarking, cross-encoder reranking, and LLM-as-a-judge evaluation - validated across 670 tests spanning unit, ML, RAG, and security tiers. Seeking an ML Engineering, AI Engineering, or MLOps post-graduation role to contribute production-grade model pipelines and LLM system design.

EDUCATION

High School of Dundee

Aug 2017 - June 2023

- National 5 Computer Science, Maths, English, Economics, Physics, Chemistry, Biology (A)
- Higher Computer Science, Maths and English (A), Higher Chemistry (B) and Higher Physics (C)
- Advanced Higher Computer Science (C)

University of Dundee - BSc (Hons) Computer Science (Data Science and AI)

Sep 2023 - Present

- Expected Graduation: June 2027

PROJECTS

ATM Log Aggregation, Analysis & Diagnostics Platform | Python, FastAPI, XGBoost, scikit-learn, MLflow, LangChain, ChromaDB, Ollama, Kafka, Redis, PostgreSQL, Docker, AWS | Mar 2026 - June 2026

- Led backend and data engineering on a 7-person Agile team for NCR Atleos; independently built the 3-layer detection engine, Agentic RAG assistant, MLOps infrastructure with MLflow experiment tracking and model registry, and a 670-test validation suite across 10 tiers - all post-submission
- Implemented a 3-layer anomaly detection engine (XGBoost + Isolation Forest, Z-score & 7 heuristic detectors) processing 2.5M+ events per day across 7 sources, engineered 49 features achieving $99.8\% \pm 0.1\%$ CV accuracy (StratifiedKFold, 5-fold) accuracy and 97.3% Isolation Forrest precision; all runs versioned in MLflow v3 on AWS (RDS backend + S3 artifact store), tracking accuracy, F1, AUC-ROC, and per-class precision/recall; promoted champion models via MLflow registry aliases, auto-retrain on artifact corruption
- Built an Agentic RAG diagnostic assistant with cross-encoder reranking (ms-marco-MiniLM-L-2-v2, +5-15% retrieval relevance), 4-stage reasoning (self-consistency scoring across 3 parallel samples, verbalised confidence, reflexion self-critique, citation grounding) and 4-signal confidence fusion (retrieval 30% + self-consistency 25% + verbalized 25% + grounding 20%); Platt scaling auto-recalibrates every 20 feedback samples, 11-23s uncached, <100ms cached, parallelising self-consistency cut latency from 27-55s to 11-23s

StockLens FinTech Spending & Investment App | React Native Expo, TypeScript, Node.js, Python, FastAPI, scikit-learn | Oct 2025 - Dec 2025

- Implemented an ML layer combining ARIMA-based time-series forecasting for per-category spending trends and Linear Regression for forward-looking portfolio projections, fed by historical CAGR computed from Alpha Vantage data - surfaced directly in the app UI alongside the OCR receipt pipeline

Haggis Species Classification & Predictive Modelling | Python, Scikit-learn, XGBoost, Matplotlib, Seaborn, Pandas, NumPy, Jupyter Notebook | Nov 2025 - Dec 2025

- Engineered and validated 2 novel ratio-based features that emerged as top-3 predictors across all models; prevented data leakage using sklearn Pipelines with ColumnTransformer and RobustScaler throughout
- Trained and compared 7 classifiers including XGBoost, Random Forest, and Logistic Regression achieving ~90% accuracy; tuned hyperparameters via GridSearchCV with 5-fold cross-validation
- Applied K-Means and DBSCAN clustering to validate natural groupings against ground-truth labels; built a Linear Regression model predicting body mass ($R^2=0.756$), validating assumptions via residual diagnostics

DevSync Project Tracker with GitHub Integration | React, Flask, PostgreSQL, Socket.io, GitHub OAuth, AWS, Docker, CI/CD | Jan 2025 - Apr 2026

- Deployed production AWS infrastructure: ECS Fargate + ALB + RDS private subnet + S3 + CloudFront + ACM; rolling deploys tagged by commit SHA for immutable rollback; OIDC federation scoped to main branch with no static credentials stored or transmitted anywhere in the pipeline, repository or task definitions
- 1,452 automated tests at 85%/85% line coverage hard-gate every PR - any failure aborts deployment before any AWS step runs, or is even triggered

PROJECTS

- CineMatch AI / ML Movie Recommendation System** | *Python, scikit-learn, Pandas, Flask, NumPy* | *May 2025 - July 2025*
- Implemented three recommendation approaches - collaborative filtering, content-based filtering, and a hybrid model - on the MovieLens dataset; collaborative filtering achieved ~78% hit rate and ~0.22 Precision@10
 - Systematically compared all three approaches across hit rate, Precision@10, and cold-start handling - hybrid model outperformed standalone methods by combining user similarity signals with item content features, with each strategy independently swappable via dependency injection for controlled benchmarking

TECHNICAL SKILLS

Languages	Python, TypeScript, JavaScript, SQL, Bash
ML & AI	Scikit-learn, XGBoost, Pandas, NumPy, MLflow
LLM Engineering	LangChain, ChromaDB, Ollama
Data & Backend	Kafka, FastAPI, Flask
Databases	SQLite, PostgreSQL, MySQL, MongoDB, Redis
Cloud & Tools	AWS (EC2, S3, RDS, IAM, SageMaker), Docker, GitHub Actions, Linux
Visualisation	Jupyter Notebook, Matplotlib, Seaborn, Power BI
Testing	Pytest

CERTIFICATES

<u>AWS Academy Graduate - Machine Learning Foundations</u>	<i>Feb 2025</i>
<u>AWS Academy Graduate - Cloud Foundations</u>	<i>Mar 2024</i>
<u>Foundations of Azure AI: Concepts, Capabilities, and Implementation</u>	<i>Oct 2024</i>

EXPERIENCE

Beaully Mini Market , Dundee, Scotland <i>Retail Assistant</i>	<i>September 2023 - Present</i>
Diagnosed and resolved Point-of-Sale and stock system failures rapidly to minimise service disruption	
Zulus Peri Peri Grill , Dundee, Scotland <i>Front of House Worker</i>	<i>June 2021 - September 2023</i>
Trained 5 new staff members; managed competing priorities across dine-in, takeaway, and delivery orders under sustained pressure	

TRANSFERRABLE SKILLS

Model Evaluation & Benchmarking Benchmark models rigorously - cross-validation, leakage prevention, held-out evaluation sets, and LLM-as-judge pipelines - not just headline accuracy scores	Experimental & Hypothesis-Driven Thinking Frame ML problems as falsifiable hypotheses; validate with held-out sets and residual diagnostics before drawing conclusions
LLM Integration & RAG Architecture Design and evaluate end-to-end LLM-powered systems - chunking strategy, embedding selection, vector retrieval, and generation quality as first-class architectural concerns	Data Pipeline Engineering for ML Build production-grade data pipelines that feed ML systems reliably - ingestion, transformation, synthetic data generation, and anomaly injection
Continuous Professional Development Independently acquired AWS ML and Azure AI certifications alongside full-time study; consistently extends technical depth beyond coursework	Research Methodology Apply structured research practices, controlled evaluation, feature validation, and statistical testing, to ensure findings generalise beyond training data
Technical Communication Present model results, retrieval quality metrics, and system trade-offs clearly to both technical peers and non-technical stakeholders	Cross-functional Collaboration Delivered ML and AI components within a multi-disciplinary Agile team; experienced coordinating across data engineering, frontend, and analysis workstreams